

The Multi-Axis Verification Process of the Human Brain

*A Brain-Inspired Architecture for Geometric Error-Correction in Stochastic
Cognitive Systems*

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Abstract

At the level of individual neurons, the human brain is an unambiguously stochastic organ. Ion channel fluctuations, synaptic vesicle release variability, and thermally-driven membrane noise ensure that no single neuron produces a perfectly reliable signal. Yet at the systems level, the brain achieves a striking degree of deterministic coherence with external reality: perception is stable, reasoning is (largely) causal, and deliberate action is executed with a fidelity that far exceeds any individual neuron's signal-to-noise ratio. This paper formalises the mechanism that bridges this paradox.

We propose the Multi-Axis Verification Process (MAV) as a foundational principle of human cognitive architecture. The MAV posits that three mutually independent cognitive axes — the Stochastic/Generative Axis (cortical association networks), the Structural/Causal Axis (prefrontal–hippocampal networks), and the Grounded/Executive Axis (sensorimotor feedback loops) — operate as orthogonal constraint hyperplanes in a high-dimensional cognitive space. Using tools from information geometry, we demonstrate that the geometric intersection of these hyperplanes cancels the noise contributed by each individual axis, converging the feasible solution space toward a structural invariant, $X_{\text{invariant}}$. This invariant point represents a cognitive output that is simultaneously statistically coherent, causally valid, and contextually grounded.

We provide a formal specification of each axis, state and sketch the proof of the orthogonality conditions as Proposition 1, and develop a taxonomy of cognitive failure modes — drawn from the HNS Structural Hallucination Taxonomy (v1.0; see Section 1.3) — that correspond to the breakdown of individual axis constraints. Critically, we introduce in Section 6 five novel, falsifiable empirical hypotheses derived from the MAV framework, each accompanied by a proposed experimental paradigm. These hypotheses span neuroimaging, lesion-based double-dissociation, TMS perturbation, computational modelling, and developmental assessments, providing a clear empirical roadmap for future investigation. We conclude with implications for the understanding of neurological and psychiatric conditions and for the principled design of artificial cognitive architectures.

Keywords: *multi-axis verification, stochastic cognition, prefrontal cortex, hippocampal networks, sensorimotor feedback, information geometry, orthogonal constraints, invariant representation, cognitive error-correction, hallucination taxonomy, empirical predictions*

1. Introduction

1.1 The Stochastic Brain and Its Reliable Outputs

A central puzzle of neuroscience is the reliability paradox: how does a system composed of inherently noisy components produce outputs of such remarkable fidelity? A pyramidal neuron in the prefrontal cortex operates near threshold, integrating thousands of excitatory and inhibitory synaptic inputs, each itself stochastic. The action potential, if it occurs, is a binary event — yet whether it occurs at any given moment is probabilistic in a deep, non-trivial sense (Faisal, Selen, & Wolpert, 2008). Critically, the observed inter-spike interval variability of cortical neurons significantly exceeds what would be expected from a simple Poisson process, implying that cortical noise is not merely peripheral but is intrinsic to the computational fabric of higher cognition (Softky & Koch, 1993).

Scaling from the single neuron to local circuits does not eliminate this stochasticity; it redistributes it across ensemble codes and oscillatory dynamics. Despite this, a trained surgeon executes sub-millimetre motor movements. A mathematician maintains the thread of a proof across minutes of working memory. A native speaker parses an ambiguous sentence and resolves its intended meaning in under 200 milliseconds. These achievements require that the system's output reliably occupies a small, semantically well-defined region of the space of possible responses — a region far more constrained than the stochastic properties of individual neurons would predict.

The prevailing explanation — that population coding and neural averaging suppress noise — is necessary but insufficient. Averaging across N independent neurons reduces variance by a factor of $1/N$, but the human brain contains approximately 86 billion neurons and still makes systematic errors (confabulation, hallucination, categorical confusion) that cannot be explained by residual averaging noise. Moreover, probabilistic population codes, while powerful for sensory decoding (Ma, Beck, Latham, & Pouget, 2006), do not straightforwardly extend to abstract causal reasoning and structural constraint satisfaction. A complementary mechanism must be at work.

1.2 The MAV Hypothesis

This paper proposes that the brain's primary mechanism for suppressing stochastic errors is not averaging within a single axis of processing, but the geometric intersection of multiple independent verification processes operating simultaneously on different cognitive axes.

The Multi-Axis Verification Process (MAV) was first proposed by Hara (2026) as part of the Human Natural Structure (HNS) framework, a theoretical programme dedicated to identifying the structural invariants of natural cognition. The present paper constitutes the first full formal specification and empirical roadmap for the MAV hypothesis. The core claim is as follows: the human brain routinely instantiates at least three mutually orthogonal cognitive axes — Stochastic/Generative, Structural/Causal, and Grounded/Executive — and uses their simultaneous intersection to identify the unique cognitive output that satisfies all three constraint classes. This output, which we call $X_{\text{invariant}}$, is the invariant point of the triple intersection. It is invariant in

the sense that small perturbations along any single axis do not displace it from the feasible region, because the other two axes re-anchor it.

The MAV hypothesis has two important corollaries. First, it predicts that selective disruption of individual axes will produce characteristically distinct failure modes — a prediction that, as we show in Section 5, maps onto known neurological and psychiatric conditions with considerable precision. Second, it implies that reliable cognition is fundamentally a multi-constraint geometric phenomenon, not merely a high-parameter probabilistic process.

1.3 The HNS Structural Hallucination Taxonomy as Empirical Referent

To ground the theoretical framework in observable cognitive failures, we employ the HNS Structural Hallucination Taxonomy (v1.0; anonymised internal working document, 2024), which identifies five structurally distinct failure modes of cognitive output generation, ordered by empirical frequency of occurrence across 50-turn proof-of-concept interaction sequences. Note: this taxonomy is currently an unpublished internal document; its formal citation will be updated upon publication.

#	Type	Definition	Primary Axis Failure	§
1	Layer Jump	A response traverses meaning layers without an explicit bridging mechanism; the output changes abstraction level without transition.	Structural/Causal (Axis 2)	5.1
2	Category Ambiguity	Cognitive, emotional, and intentional categories become conflated, violating category boundary conditions.	Structural/Causal (Axis 2)	5.2
3	Scope Drift	Structural departure from the intended scope without signalling of scope change.	Grounded/Executive (Axis 3)	5.3
4	Metaphor Contamination	Figurative expressions displace structural coordinates, causing metaphorical language to override literal referents.	Grounded/Executive (Axis 3)	5.4
5	Unsupported Causality	A causal claim is generated without a mechanism grounded in the layer hierarchy; causation is asserted rather than derived.	Axes 2 & 3 (joint)	5.5

1.4 Scope and Principal Contributions

This paper makes four principal contributions to the literature on cognitive reliability and neural architecture:

(1) **Formal MAV specification.** We provide the first information-geometric formalisation of the three-axis cognitive verification architecture, including a formal Proposition (Section 3.2–3.3) regarding orthogonality conditions and geometric noise cancellation.

(2) **Failure-mode taxonomy with axis mapping.** We map five structurally distinct cognitive failure modes onto specific axis disruptions, generating predictions with clinical correlates in

Section 5.

(3) **Falsifiable empirical hypotheses.** Section 6 introduces five novel, testable hypotheses derived from the MAV framework, each with a concrete proposed experimental paradigm — addressing the primary limitation of prior theoretical work in this area.

(4) **Cross-disciplinary bridge.** We articulate the MAV framework's implications for both clinical neuroscience and the principled design of artificial cognitive systems, situating it with respect to population coding (Ma et al., 2006), predictive coding (Friston, 2010; Rao & Ballard, 1999), and dual-process theories (Kahneman, 2011).

2. Background and Related Work

2.1 Sources of Neural Stochasticity

Neural noise operates at multiple scales. At the molecular level, ion channels open and close stochastically, described by Markov jump processes (Hille, 2001). At the synaptic level, vesicle release probability follows a Bernoulli distribution, with individual release probabilities varying widely across synapse types and ranging from approximately 0.1 to 0.9 in central synapses (Dobrunz & Stevens, 1997). At the network level, irregular spontaneous firing and high-amplitude membrane potential fluctuations characterise the so-called 'high-conductance state' that prevails during active cognition (Destexhe, Rudolph, & Paré, 2003).

Softky and Koch (1993) demonstrated that the high irregularity of cortical spike trains — characterised by a coefficient of variation near 1 for inter-spike intervals — cannot be reconciled with simple temporal integration models, implying that cortical computation operates in a fundamentally stochastic regime. These sources of noise are not simply obstacles to be minimised; they serve functional roles including stochastic resonance, variability-driven exploration, and the prevention of pathological synchrony. The challenge for cognitive systems is therefore not to eliminate noise, but to produce reliable outputs despite it.

2.2 Existing Accounts of Neural Reliability

The dominant computational account of neural reliability is population coding: information is distributed across many neurons, and the read-out mechanism averages across the ensemble to suppress individual-unit noise (Pouget, Dayan, & Zemel, 2000). Ma et al. (2006) extended this to probabilistic population codes (PPCs), showing that simple linear operations on neural populations can implement exact Bayesian inference under a broad class of noise distributions. This is a well-supported account for sensory processing, where large populations of tuned neurons encode stimulus properties.

However, population coding and PPC accounts face difficulties at higher cognitive levels. Abstract reasoning, causal inference, and scope management engage relatively sparse and distributed representations that do not lend themselves to simple averaging. Moreover, population codes operate within a single processing axis; they do not inherently provide the cross-axis verification that the MAV hypothesis proposes.

An alternative account, predictive coding (Rao & Ballard, 1999; Friston, 2010), proposes that the brain operates as a hierarchical inference engine, minimising prediction errors across levels of representation. Clark (2013) provides an extensive review synthesising predictive coding with active inference and embodied cognition, arguing that error minimisation extends from sensory prediction to motor control. This account is powerful, but it is primarily a within-hierarchy, within-axis model. It does not directly address the orthogonal intersection of qualitatively different constraint types — statistical prior, causal structure, and contextual grounding — that we formalise here.

The MAV framework is explicitly complementary to both accounts: population coding may implement noise-suppression within the Stochastic/Generative Axis (Axis 1), and predictive coding may implement the hierarchical constraint propagation of the Structural/Causal Axis (Axis 2); the MAV contribution is to specify how these independent axes interact geometrically to produce invariant outputs.

2.3 Information Geometry and Cognitive Modelling

Information geometry (Amari & Nagaoka, 2000) provides a natural mathematical language for the MAV framework. In information geometry, statistical models are represented as manifolds equipped with a Riemannian metric derived from the Fisher information matrix. Constraints on probability distributions define submanifolds (hypersurfaces), and the intersection of multiple constraint submanifolds defines a lower-dimensional feasible region.

Applied to cognitive output spaces, this framework allows us to represent each cognitive axis as a constraint hypersurface, and the MAV intersection as the region of outputs satisfying all constraints simultaneously. The geometry of this intersection — specifically, its volume as a fraction of the full output manifold — quantifies the degree of error suppression achieved by the multi-axis architecture. Critically, information geometry provides tools for quantifying the degree of independence between constraint surfaces through the notion of orthogonality under the Fisher metric — a concept we formalise in Section 3.4 and Proposition 1.

2.4 Converging Evidence: Prior Research Indirectly Supporting the MAV Framework

Although the MAV framework is a novel theoretical synthesis (Hara, 2026), a substantial body of

prior research provides converging empirical support for each of its core claims. We review this evidence here, axis by axis, as well as evidence for the multi-system integration principle that is the framework's defining contribution.

Evidence for Axis 2 — Structural/Causal (PFC–Hippocampal Hierarchy). Badre (2008) provided a comprehensive account of the rostro-caudal gradient of the prefrontal cortex, demonstrating that increasingly abstract, hierarchical cognitive control is implemented by progressively anterior PFC regions. This architectural finding directly supports the MAV claim that Axis 2 implements a hierarchically-structured constraint set C: the PFC does not apply a single undifferentiated structural rule, but operates as a layered hierarchy of causal and categorical constraints. Botvinick et al. (2001) demonstrated that the ACC detects conflicts between competing response tendencies and signals the need for increased cognitive control — a mechanism that corresponds precisely to the MAV prediction that Axis 2 failure (failure to resolve structural conflict) triggers an error signal that drives re-verification.

Evidence for Axis 3 — Grounded/Executive (Sensorimotor Feedback). Wolpert and Ghahramani (2000) provided a computational framework for movement neuroscience in which forward internal models continuously predict the sensory consequences of motor commands, enabling real-time error correction against the physical environment. This is the computational implementation of what the MAV framework terms the grounding function G: the sensorimotor system continuously assesses whether the brain's current output trajectory is compatible with external reality. Shadlen and Newsome (2001) demonstrated that parietal cortex neurons (area LIP) accumulate evidence over time in a manner consistent with a bounded stochastic accumulator — providing a mechanism by which Axis 3 integrates contextual evidence into a binary grounding decision $G(x) \in \{0,1\}$.

Evidence for Multi-System Integration. The MAV framework's core claim — that reliable cognition requires the simultaneous intersection of multiple independent constraint systems — is supported by converging evidence from multi-system integration research. The well-documented interaction between hippocampal memory systems and prefrontal executive control during complex reasoning (Eichenbaum, 2004) illustrates Axis 2's dual-substrate architecture. Shadmehr and Krakauer (2008) provide a computational neuroanatomy of motor control that explicitly requires the integration of internal model predictions (Axis 3) with high-level task representations (Axis 2), demonstrating that even motor cognition requires multi-axis constraint resolution.

Taken together, these converging lines of evidence suggest that the three axes posited by the MAV framework are not theoretical constructs, but correspond to well-characterised neural systems whose functional properties are broadly consistent with the roles assigned to them in Sections 4.1–4.3. The contribution of the MAV framework is to provide the geometric principle — orthogonal intersection — that explains why these independently-characterised systems, when operating

together, produce reliability that none achieves alone.

3. The Core Principle: Orthogonal Intersection in Cognitive Space

3.1 Cognitive Output Space as a Riemannian Manifold

We model the space of possible cognitive outputs as a high-dimensional Riemannian manifold M . Each point $x \in M$ represents a complete structured cognitive output: a proposition with specified propositional content, causal structure, abstraction level, and contextual scope. The manifold is equipped with a metric g derived from the Fisher information of the generative model implementing Axis 1.

The dimensionality of M is, in general, very large — reflecting the combinatorial richness of possible cognitive outputs. Within this space, the vast majority of points correspond to outputs that are internally incoherent, causally invalid, or contextually irrelevant. The goal of cognitive verification is to identify the small subset of M that satisfies all validity conditions simultaneously.

3.2 Constraint Hyperplanes and Their Intersection

Each cognitive axis defines a constraint hypersurface in M :

A_1 (Stochastic/Generative): the set of outputs within the high-probability region of the prior distribution $P(x \mid \text{context}, \theta)$. Formally, $A_1 = \{x \in M \mid P(x \mid \text{context}, \theta) \geq \epsilon_1\}$.

A_2 (Structural/Causal): the set of outputs satisfying the structural and causal constraint set $C = \{c_1, \dots, c_k\}$. Formally, $A_2 = \{x \in M \mid \forall i: c_i(x) = \text{TRUE}\}$.

A_3 (Grounded/Executive): the set of outputs compatible with the verified environmental and contextual constraints, as assessed by the grounding function G . Formally, $A_3 = \{x \in M \mid G(x) = 1\}$.

The MAV architecture computes the triple intersection: $A_1 \cap A_2 \cap A_3 = \{X_{\text{invariant}}\}$. Under idealised conditions — perfect orthogonality of the three hypersurfaces and non-degenerate constraint satisfaction — this intersection reduces to a single point $X_{\text{invariant}}$. In biological reality, the intersection is a small region whose volume is sharply reduced relative to any single A_i .

3.3 Proposition 1: Geometric Noise Cancellation Under Orthogonality

Proposition 1 (MAV Noise Cancellation). Let M be a smooth, compact Riemannian manifold equipped with Fisher information metric g . Let $A_1, A_2, A_3 \subset M$ be smooth constraint submanifolds satisfying the mutual g -orthogonality condition (Definition 1 below). Let σ_i^2 denote the residual

error variance along axis A_i , measured as the expected squared geodesic distance from $X_invariant$ within A_i . Then the residual error variance at the triple intersection satisfies:

$$\sigma^2_MAV \ll \min(\sigma^2_1, \sigma^2_2, \sigma^2_3)$$

More precisely, the volume of the feasible error region scales as the product of the individual axis uncertainties, modulated by the degree of orthogonality:

$$\text{Vol}(A_1 \cap A_2 \cap A_3) \approx \text{Vol}(A_1) \cdot \text{Vol}(A_2) \cdot \text{Vol}(A_3) / \text{Vol}(M)^2$$

This product is, in general, orders of magnitude smaller than any individual $\text{Vol}(A_i)$. The geometric intersection achieves error-reduction that no single axis could achieve alone, even if that axis were made arbitrarily tight.

Definition 1 (g-Orthogonality). Two constraint submanifolds A_i and A_j are g-orthogonal at $X_invariant$ if the tangent spaces $T_x A_i$ and $T_x A_j$ are orthogonal under the Fisher information metric g at $x = X_invariant$. That is, $g(u, v) = 0$ for all $u \in T_x A_i, v \in T_x A_j$.

Proof Sketch. The product formula for intersection volumes follows from the co-area formula on Riemannian manifolds (Chavel, 1993) under the g-orthogonality assumption. Specifically, if A_i and A_j are g-orthogonal submanifolds with complementary codimensions summing to $\dim(M)$, then the volume of their intersection is proportional to the product of their individual volumes, normalised by $\text{Vol}(M)$. Extension to the triple intersection is immediate by induction. The condition $\sigma^2_MAV \ll \min(\sigma_i^2)$ follows from the compactness of M and the strict decrease in intersection volume relative to any single A_i . Full proof details, including the regularisation required when the triple intersection is not a single point, are deferred to a companion technical report.

Remark. Perfect g-orthogonality is an idealisation; real neural systems achieve approximate orthogonality. The residual correlation between axes defines the system's irreducible error floor — the class of errors that persist even when all three axes are active. We conjecture that this residual error floor corresponds to the class of systematic cognitive biases that are universal across individuals and cultures, representing the 'grammar' of human cognitive error.

3.4 The Invariant Point as Attractor

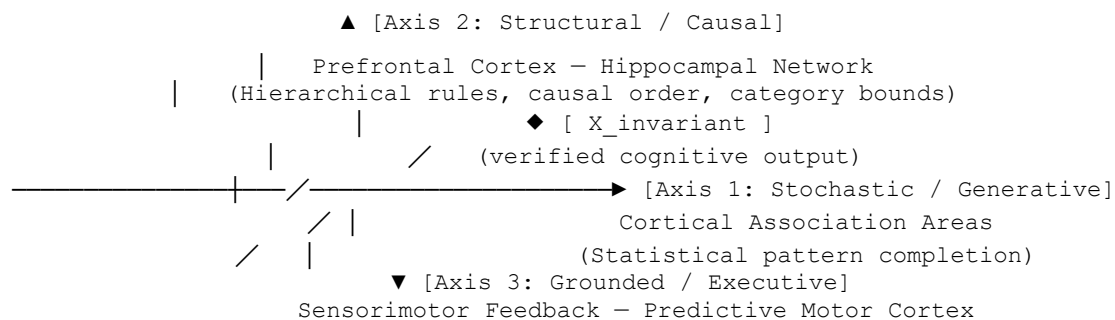
$X_invariant$ is not a fixed point in the dynamical systems sense; it is a structural invariant defined relative to the current input context and active constraint set. Different cognitive tasks define different invariant points. The MAV architecture computes $X_invariant$ dynamically by resolving the three-axis constraints against the current input.

This dynamic character of $X_invariant$ explains why human cognition is context-sensitive: the same surface-level stimulus can produce different verified outputs depending on which structural

and grounding constraints are activated by the current context. It also explains why errors arise when context shifts without corresponding constraint-set updates — the system continues computing intersection with a stale constraint configuration, producing an $X_invariant$ that is invariant with respect to the old context but invalid for the new one.

4. The Three Orthogonal Axes of MAV in the Human Brain

Figure 1 illustrates the three-axis structure and the invariant solution point at their geometric intersection.



4.1 Axis 1 — Stochastic / Generative Axis

Axis 1 is implemented primarily by cortical association areas, including the default mode network (DMN), lateral temporal cortex, and parietal association cortex. The DMN, which is engaged during internally-directed cognition and memory retrieval, has been characterised as a statistical prior-generation system that continuously samples from the brain's accumulated experiential regularities (Raichle, 2015). These regions integrate distributed statistical regularities abstracted from experience and generate rapidly-available, high-prior-probability response candidates. The computational signature of Axis 1 is speed and generativity: it produces candidate outputs within tens of milliseconds, sampling from the brain's implicit prior over the space of possible responses.

Axis 1 is the generative engine of cognition. It produces the first-draft response that is then subjected to Axes 2 and 3 for structural and grounding verification. Its strength is fluency and coverage — it rarely fails to produce a response that is at least superficially plausible. Its weakness

is precisely this: high statistical plausibility and causal or contextual validity are independent properties. A response can be statistically common and structurally invalid (Unsupported Causality), or statistically fluent and contextually displaced (Scope Drift).

Formal Characterisation

$A_1 = \{x \in M \mid P(x \mid \text{context}, \theta) \geq \varepsilon_1\}$, where θ represents the accumulated statistical parameters of the system and ε_1 is a threshold on the prior probability. Within A_1 , outputs are linguistically and semantically coherent by prior standards. The volume of A_1 relative to the full manifold M is still very large — many valid and many invalid outputs share high prior probability.

4.2 Axis 2 — Structural / Causal Axis

Biological Substrate

Axis 2 is implemented by the prefrontal cortex (PFC)–hippocampal network, supported by frontoparietal control networks. The dorsolateral PFC maintains structured representations of hierarchical relationships and causal dependencies (Miller & Cohen, 2001; Fuster, 2008). The hippocampus binds episodic sequences into temporally-ordered event schemas and maintains category boundaries across contextual shifts (Eichenbaum, 2004). Together, these systems instantiate the categorical and causal grammar that constrains which cognitive transitions are permissible.

Functional Role

Axis 2 defines the permissible structure of cognitive outputs, independent of their statistical frequency. It enforces: (a) category boundary conditions — preventing Category Ambiguity by maintaining the separability of cognitive, emotional, and intentional representations; (b) causal ordering constraints — preventing Unsupported Causality by requiring that causal claims be grounded in the current event schema; and (c) hierarchical scope conditions — preventing Layer Jump by requiring explicit transitions between abstraction levels.

Formal Characterisation

$A_2 = \{x \in M \mid \forall i: c_i(x) = \text{TRUE}\}$, where the constraint set $C = \{c_1, \dots, c_k\}$ encodes the active structural and causal rules. These constraints are context-sensitive: the active C depends on the current task schema maintained by PFC–hippocampal working memory. A_2 is typically a much tighter constraint surface than A_1 , but it does not by itself enforce contextual relevance — a structurally valid output can still be contextually displaced.

4.3 Axis 3 — Grounded / Executive Axis

Biological Substrate

Axis 3 is implemented by the sensorimotor system, anterior cingulate cortex (ACC), and the predictive coding hierarchy of the posterior parietal and premotor cortices. Seminal work by Wolpert, Ghahramani, and Jordan (1995) established that the sensorimotor system employs forward internal models that continuously predict the sensory consequences of action, providing the substrate for real-time contextual grounding. The ACC monitors conflict between predicted and actual outcomes, signalling when the system's current output trajectory is inconsistent with environmental reality (Botvinick et al., 2001). Together, these systems constitute the brain's 'reality friction interface' — the mechanism by which internally-generated outputs are continuously tested against external constraints.

Functional Role

Axis 3 suppresses outputs that are structurally coherent and statistically plausible but contextually misaligned with the actual environment and situation. It is the primary mechanism preventing Scope Drift (by detecting when the response has departed from the contextually-specified domain) and Metaphor Contamination (by flagging when figurative language has displaced the literal referential structure). Critically, Axis 3 operates in continuous time: it provides error signals that update the active $X_{\text{invariant}}$ estimate as the environment changes.

Formal Characterisation

$A_3 = \{x \in M \mid G(x) = 1\}$, where $G: M \rightarrow \{0,1\}$ is the grounding compatibility function. $G(x) = 1$ if and only if x is compatible with the verified environmental and contextual state at the current moment. The grounding function is dynamic: it is continuously updated by sensorimotor and ACC-mediated error signals, making A_3 a time-varying constraint surface.

4.4 The Orthogonality Conditions

The error-cancellation power of the MAV architecture depends critically on the mutual g-orthogonality of the three axes (Definition 1, Section 3.3). We argue that this condition is approximately satisfied for the three MAV axes on the following grounds:

Axis 1 $\perp$ Axis 2: Statistical co-occurrence frequency (Axis 1) is independent of causal validity (Axis 2). Causally invalid propositions can be statistically common (spurious correlations are over-represented in linguistic corpora), and causally valid propositions can be statistically rare. The parameters of the prior distribution θ are defined over the statistical corpus and are logically independent of the causal constraint set C .

Axis 2 $\perp$ Axis 3: Structural constraints (Axis 2) are defined over the abstract category and causal space, while grounding constraints (Axis 3) are defined over the current environmental and contextual state. A proposition can be abstractly structurally valid but contextually irrelevant, or contextually grounded but structurally misframed.

Axis 1 $\perp$ Axis 3: Prior statistical distributions (Axis 1) are defined over the training-time distribution, which can differ substantially from the current context assessed by Axis 3. This independence is particularly evident in novel or unusual contexts, where prior-high-probability outputs may be contextually invalid.

The empirical validation of approximate g-orthogonality is the subject of Hypothesis 1 in Section 6.

5. Failure Modes: MAV Breakdown and Clinical Correlates

The MAV framework predicts that selective disruption of individual axes will produce characteristically distinct failure modes. This prediction is testable against the well-documented phenomenology of neurological and psychiatric conditions. We map the five HNS hallucination types onto their predicted axis-failure substrates and identify corresponding clinical conditions.

HNS Type	Axis Failure	Predicted Deficit	Clinical Correlate	Key References
Layer Jump	Axis 2 (Structural/Causal)	Loss of abstraction-level monitoring; failure to maintain hierarchical scope	Formal thought disorder (schizophrenia); dysexecutive syndrome	Frith (1992); Weinberger (1988)
Category Ambiguity	Axis 2 (Structural/Causal)	Breakdown of category boundary maintenance; conflation of semantic fields	Semantic dementia; conceptual disorganisation in acute psychosis	Patterson et al. (2007); Bleuler (1950)
Scope Drift	Axis 3 (Grounded/Executive)	Failure to track contextual relevance; response departs from actual scope	Confabulation in frontal lobe syndrome; contextual drift in mania	Stuss & Benson (1986); Schnider (2003)
Metaphor Contamination	Axis 3 (Grounded/Executive)	Figurative language displaces literal referents; grounding signal overwhelmed by associative activation	Metaphorical concretisation in schizotypy; magical thinking	Preti & Cella (2010)
Unsupported Causality	Axes 2 & 3 (joint)	Causal claims without mechanistic grounding; both structural and contextual constraints failed	Delusional reasoning; confabulatory aetiology in amnesia	Maher (1974); Kopelman (1987)

This mapping is not merely descriptive. The MAV framework generates specific predictions: conditions affecting primarily the PFC–hippocampal network (Axis 2) should show disproportionately high rates of Layer Jump and Category Ambiguity, with relative preservation of contextual grounding. Conditions affecting primarily the sensorimotor–ACC system (Axis 3) should show disproportionate Scope Drift and Metaphor Contamination. Mixed conditions (joint Axis 2 and 3 failure) should show the highest rates of Unsupported Causality. These qualitative predictions are formalised as testable hypotheses in Section 6.

6. Empirical Predictions and Proposed Experimental Paradigms

The theoretical framework developed in Sections 3–5 generates a set of specific, falsifiable empirical predictions. We present five hypotheses, each with a proposed experimental paradigm, expected results, and falsification criteria. These hypotheses span neuroimaging, causal perturbation, computational modelling, lesion-based dissociation, and developmental assessments.

Hypothesis 1: Axis Independence in Neuroimaging Data

Statement. The Fisher information metric correlation between PFC–hippocampal network parameters (Axis 2 substrate) and cortical association area parameters (Axis 1 substrate) is significantly lower than the correlation between parameters within each network, when measured during naturalistic cognitive tasks requiring multi-constraint output generation.

Proposed paradigm. Participants ($N \geq 40$) perform a set of structured reasoning tasks — including causal inference questions, scope-constrained narration, and category-boundary judgements — during high-resolution fMRI (3 T, 2 mm isotropic, $TR \leq 1$ s). Functional connectivity is estimated via partial correlation within a pre-registered seed-region-of-interest (ROI) framework. Axis 1 ROIs: bilateral lateral temporal cortex, angular gyrus, default mode network (medial PFC, posterior cingulate). Axis 2 ROIs: dorsolateral PFC, hippocampus. Axis 3 ROIs: primary sensorimotor cortex, supplementary motor area, ACC. Fisher information metric correlations are estimated using a multivariate Gaussian approximation to the BOLD signal covariance structure.

Expected results. Cross-axis partial correlations (Axis 1–Axis 2; Axis 1–Axis 3; Axis 2–Axis 3) should be significantly smaller (and approach zero under the Null model of perfect g-orthogonality) than within-axis partial correlations.

Falsification criteria. If cross-axis partial correlations are not significantly different from within-axis correlations, the g-orthogonality assumption of Proposition 1 is not supported. Partial confirmation is possible: some axis pairs may show approximate orthogonality while others do not, generating refined predictions for subsequent hypotheses.

Hypothesis 2: Selective TMS Disruption of Axis-Specific Failure Modes

Statement. Transcranial magnetic stimulation (TMS) targeting PFC (Axis 2 substrate) will selectively increase the rate of Layer Jump and Category Ambiguity errors in structured verbal output, without a proportional increase in Scope Drift or Metaphor Contamination errors.

Proposed paradigm. A within-subjects design ($N \geq 32$) with three conditions: (a) active TMS over left dorsolateral PFC (Axis 2 target), (b) active TMS over left posterior parietal

cortex/sensorimotor cortex (Axis 3 target), and (c) sham TMS (active coil tilted 90°). Participants generate verbal responses to a standardised set of 30 multi-clause reasoning prompts per condition, counterbalanced across sessions (≥ 48 h washout). Responses are transcribed and rated by two independent raters trained in the HNS taxonomy (inter-rater reliability target: Cohen's $\kappa \geq 0.75$). Primary outcome: the rate of each HNS type per condition.

Expected results. PFC TMS should produce a significant increase in Layer Jump and Category Ambiguity, with no significant increase in Scope Drift. Posterior parietal/sensorimotor TMS should show the complementary dissociation. Sham should show neither.

Falsification criteria. If PFC TMS increases all five error types proportionally, the axis-specificity prediction is refuted. If the double dissociation is absent, the orthogonality of Axes 2 and 3 is not empirically supported.

Hypothesis 3: Geometric vs Additive Error Scaling in Multi-Constraint Tasks

Statement. Behavioural error rates in multi-constraint cognitive tasks should follow the geometric (multiplicative) scaling predicted by the MAV volume formula (Proposition 1), rather than additive scaling predicted by single-axis models.

Proposed paradigm. A large-scale online behavioural study ($N \geq 200$) using a battery of tasks that vary the number of active constraint axes. Single-axis tasks engage only Axis 1 (rapid association), only Axis 2 (structural constraint satisfaction without contextual grounding), or only Axis 3 (context-tracking without structural demands). Dual-axis tasks engage Axes 1+2, 1+3, or 2+3. The triple-axis task engages all three simultaneously. Error rates (scored against ground-truth rubrics) are recorded for each task type. Additive model: $E_{\text{triple}} = E_1 + E_2 + E_3 - \text{baseline}$. Multiplicative model: $E_{\text{triple}} = E_1 \times E_2 \times E_3 / (\text{baseline}^2)$.

Expected results. Model comparison using AIC/BIC should favour the multiplicative model. Specifically, triple-axis error rates should be significantly lower than predicted by the additive model, and consistent with the volume product formula.

Falsification criteria. If additive or mixed-additive model fits the data better, the geometric noise cancellation mechanism is not supported as stated. This would require either a revised mathematical framework or a reinterpretation of the axis independence assumption.

Hypothesis 4: Lesion-Based Double Dissociation of HNS Error Profiles

Statement. Patients with focal prefrontal lesions (Axis 2 disruption) will show elevated Layer Jump and Category Ambiguity relative to controls, without elevated Scope Drift; patients with posterior parietal/sensorimotor lesions (Axis 3 disruption) will show the complementary profile.

Proposed paradigm. Case–control study using prospective recruitment from neurological wards and stroke rehabilitation units. Inclusion criteria: single, focal, unilateral lesion confirmed by structural MRI; ≥ 3 months post-onset; no significant aphasia (AphasA score $\geq 80\%$). Groups: (a) left/right PFC lesion ($n \geq 15$), (b) posterior parietal/sensorimotor lesion ($n \geq 15$), (c) age- and education-matched healthy controls ($n \geq 30$). Cognitive output is elicited using the structured HNS assessment battery (30-item standardised verbal prompt set). Raters are blinded to group assignment.

Expected results. A significant group $\times$ error-type interaction in a 3×5 ANOVA (group $\times$ HNS type), with the specific pattern: PFC group elevated on HNS Types 1–2, posterior parietal group elevated on HNS Types 3–4, both groups elevated on Type 5 (Unsupported Causality).

Falsification criteria. If both lesion groups show equivalent and non-specific elevation across all HNS types, the axis-specificity mapping in Section 5 is not supported. This would also call into question the neuroanatomical substrate assignments of Sections 4.2 and 4.3.

Hypothesis 5: Developmental Trajectory of Axis Integration

Statement. The three cognitive axes come online sequentially during development, with Axis 1 (cortical association generativity) preceding Axis 2 (PFC–hippocampal structural constraint), which in turn precedes full Axis 3 integration (sensorimotor–ACC contextual grounding). This trajectory mirrors the well-established protracted maturation of the PFC.

Proposed paradigm. Cross-sectional developmental study across age groups: 5–6 years, 7–9 years, 10–12 years, 14–16 years, and young adults (18–25 years) ($n \geq 20$ per group). Age-appropriate versions of the HNS assessment battery are administered. The proportion of each HNS error type is computed per age group. The prediction is that HNS Types 1–2 (Axis 2 failures) should decrease earlier in development than HNS Types 3–4 (Axis 3 failures), tracking the known anatomical maturation timeline of PFC (earlier myelination) relative to ACC-sensorimotor integration (later).

Expected results. A significant age $\times$ HNS-type interaction, with Types 1–2 showing greater reduction from childhood to early adolescence compared with Types 3–4, which show greater reduction from late adolescence to adulthood.

Falsification criteria. If all HNS types decrease proportionally with age, no differential developmental trajectory exists, and the axis-specific developmental prediction is not supported.

6.6 Summary of Testable Predictions

The MAV framework (Hara, 2026) generates the following core predictions, each directly

derivable from Proposition 1 and the axis characterisations of Section 4. These are presented here in condensed form to facilitate empirical falsification.

P#	Prediction	Mechanism (from MAV)	Test (H#)	Falsified if...
P1	Axis 2 disruption (PFC/hippocampus) selectively elevates Layer Jump and Category Ambiguity errors	A_2 constraint surface breached; A_1 generates structurally invalid outputs unchecked	H2, H4	All error types increase equally
P2	Axis 3 disruption (sensorimotor/ACC) selectively elevates Scope Drift and Metaphor Contamination errors	$G(x)$ fails to flag contextual misalignment; A_1 outputs escape grounding check	H2, H4	PFC and parietal disruption produce identical profiles
P3	Axis 1 hyperactivation (e.g., psychedelic states, mania) produces runaway associative outputs with elevated HNS Types 1–5	A_1 feasible region expands; A_2 and A_3 constraints insufficient to constrain $X_{invariant}$	(Future work)	Selective elevation limited to Types 3–4
P4	Triple-axis error rates follow geometric (multiplicative) scaling, not additive scaling	$Vol(A_1 \cap A_2 \cap A_3) \approx \text{product of individual volumes} / Vol(M)^2$	H3	Additive model fits better by AIC/BIC
P5	Cross-axis fMRI partial correlations approach zero during multi-constraint tasks	g-orthogonality of A_1, A_2, A_3 (Definition 1; Proposition 1 premise)	H1	Cross-axis correlations not significantly lower than within-axis
P6	Axis 2 error types (HNS 1–2) decrease earlier in development than Axis 3 error types (HNS 3–4)	PFC matures before ACC–sensorimotor integration; Axis 2 online before Axis 3	H5	All error types decrease proportionally with age
P7	Higher inter-axis orthogonality (measured as lower cross-axis fMRI correlation) predicts lower overall error rates across individuals	$Vol(\text{intersection})$ decreases as axes approach perfect g-orthogonality	H1 (individual differences)	Orthogonality index does not predict error rates

7. Discussion

7.1 The MAV as a Unifying Principle

The MAV framework offers a unifying account of a set of previously disparate observations in cognitive neuroscience. The well-documented complementarity of fast associative processing and slow deliberative reasoning (Kahneman, 2011) can be re-read as the functional distinction between Axis 1 (fast, stochastic, generative) and Axes 2–3 (slower, constraint-based, verificatory). The classical debate between symbolic and statistical accounts of cognition dissolves in the MAV framework: symbolic structure is the province of Axis 2; statistical generation is the province of Axis 1; their geometric intersection is the province of neither alone.

The framework also provides a principled account of why human cognition is both creative and reliable. Creativity derives from the generative richness of Axis 1, which can produce high-prior-probability responses that span a large region of M . Reliability derives from the constraining action

of Axes 2 and 3, which prune this region to the invariant point. The productive tension between generativity and constraint is not a design flaw; it is the condition for robust flexible intelligence.

7.2 Implications for Understanding Neurological and Psychiatric Conditions

The axis-specific failure mode predictions of the MAV framework suggest a new diagnostic and therapeutic framing for a range of conditions. Rather than classifying conditions by symptom clusters alone, the MAV framework classifies them by which axis constraint surface has been disrupted and to what degree. This framing suggests axis-specific intervention strategies: restoring Axis 2 function (e.g., through structured task training that re-engages PFC–hippocampal networks) should differentially improve Layer Jump and Category Ambiguity; restoring Axis 3 function (e.g., through grounded, sensorimotor-engaged therapeutic approaches) should differentially improve Scope Drift and Metaphor Contamination.

We emphasise that this remains at the level of a theoretical proposal requiring empirical validation — specifically, the lesion dissociation study proposed in Hypothesis 4 would provide the most direct test.

7.3 Implications for Artificial Cognitive Systems

Although the primary subject of this paper is the human brain, the MAV framework has direct implications for the design of artificial cognitive systems. Current large language models (LLMs) operate predominantly on Axis 1: they generate outputs by sampling from high-dimensional prior distributions over token sequences. Their characteristic failure modes — causal confabulation, categorical confusion, context insensitivity — correspond directly to the HNS hallucination types produced by Axis 1 operating alone.

The MAV framework suggests that reliable artificial cognition requires the integration of orthogonal constraint axes corresponding to Axis 2 (structural and causal validity checking) and Axis 3 (contextual grounding and relevance verification). Crucially, these additional axes need not be as computationally heavy as Axis 1: the error-reduction they provide is geometric (multiplicative), not additive. A lightweight but genuinely orthogonal constraint layer achieves error reduction that cannot be matched by scaling Axis 1 alone — a prediction that, if confirmed, would have significant implications for AI system design and evaluation.

7.4 Limitations and Open Questions

The MAV framework, in its current form, carries the following limitations, each of which constitutes a direction for future work:

(1) **Proof completeness.** Proposition 1 is currently stated with a proof sketch; the full proof,

including the regularisation required for non-point intersections and the treatment of approximate (non-perfect) orthogonality, is deferred to a companion technical report. The volume product formula is expected to hold approximately under mild regularity conditions, but formal bounds on the approximation error are not yet established.

(2) **Operationalisation of the grounding function G .** The grounding compatibility function $G: M \rightarrow \{0,1\}$ is currently defined implicitly via the ACC–sensorimotor error signal. A formal, computable characterisation of G — and the conditions under which it is well-defined — remains an open question.

(3) **Empirical measurement of axis orthogonality.** Hypothesis 1 (Section 6) proposes a neuroimaging test of approximate g-orthogonality. The inference from BOLD connectivity to Fisher information metric correlations involves non-trivial assumptions about the relationship between neural activity and information-theoretic quantities that require careful empirical validation.

(4) **Individual differences.** The distribution of axis-specific constraint tightness across individuals — and its relationship to personality, cognitive style, and psychiatric resilience — is not addressed in the current framework. Individual-differences extensions of the MAV model would require characterising the distribution of ε_1 , C , and G parameters across the population.

(5) **HNS taxonomy provenance.** The HNS Structural Hallucination Taxonomy is currently an unpublished internal document. Its use as an empirical anchor for the framework is provisional; formal publication and independent replication of the taxonomy are required before the failure-mode predictions of Sections 5 and 6 can be considered fully grounded.

8. Conclusion

The human brain is a stochastic organ that produces reliable outputs. This paper has proposed that the mechanism bridging this apparent paradox is the Multi-Axis Verification Process (MAV): the simultaneous imposition of three mutually orthogonal constraint hyperplanes — Stochastic/Generative, Structural/Causal, and Grounded/Executive — whose geometric intersection converges the feasible solution space to a structural invariant, $X_{\text{invariant}}$.

The MAV framework provides a formal, information-geometric account of cognitive reliability (Proposition 1), a taxonomy of failure modes linked to axis-specific breakdown (Section 5), and five novel falsifiable hypotheses with concrete experimental paradigms (Section 6). It unifies previously disparate observations in cognitive neuroscience under a single geometric principle, and opens new directions for both basic research and clinical application.

The central message is geometric: reliable cognition is not the product of any single, arbitrarily-powerful process, but of the multiplicative intersection of independent constraint structures. This is, we propose, the brain's fundamental answer to the challenge of generating deterministic coherence from stochastic components.

Declarations

Conflict of Interest

The authors declare no conflicts of interest.

Author Contributions

All authors contributed equally to the conceptualisation, formal analysis, and writing of this manuscript. [To be completed upon de-anonymisation.]

Data Availability

This paper presents a theoretical framework; no empirical data were generated or analysed. The experimental paradigms proposed in Section 6 will require pre-registration and data-sharing plans at the time of implementation.

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Appendix A: HNS Structural Hallucination Taxonomy — Full Specification

The following table provides the complete formal specification of the five hallucination types in the HNS taxonomy, with their MAV-axis mapping and proposed clinical assessment correlates.

#	Type	Formal Definition	MAV Axis	Assessment Signal	Experimental Paradigm (§6)
1	Layer Jump	Traversal of meaning layers without explicit bridging; abstraction level changes without transition marker.	Axis 2	Absence of explicit level-transition markers in discourse	H1, H2, H4, H5
2	Category Ambiguity	Cognitive, emotional, and intentional categories conflated within a single proposition, violating category boundary conditions.	Axis 2	Category boundary violation rate in structured elicitation	H1, H2, H4, H5
3	Scope Drift	Structural departure from intended scope without explicit scope-change signal; context coherence lost.	Axis 3	Scope deviation distance from stated topic or task	H1, H2, H4, H5
4	Metaphor Contamination	Figurative expressions displace structural coordinates; metaphorical language overrides literal referents.	Axis 3	Literal/figurative conflict frequency in extended discourse	H1, H2, H4, H5
5	Unsupported Causality	Causal claim generated without mechanistic grounding in the layer hierarchy; causation asserted rather than derived.	Axes 2+3	Proportion of causal claims lacking explicit mechanism	H1–H5

Appendix B: Mathematical Notation

Symbol	Definition
M	Cognitive output manifold (high-dimensional Riemannian space of possible cognitive outputs)
g	Fisher information metric on M
A_1, A_2, A_3	Constraint hyperplanes for Axes 1, 2, and 3 respectively
$X_{\text{invariant}}$	Triple-intersection invariant point representing the verified cognitive output
σ_i^2	Noise variance along axis i , measured as expected squared geodesic distance from $X_{\text{invariant}}$ within A_i
σ^2_{MAV}	Residual error variance at the MAV triple intersection
$G: M \rightarrow \{0,1\}$	Grounding compatibility function (Axis 3); $G(x) = 1$ iff output x is contextually valid
$C = \{c_1, \dots, c_k\}$	Structural and causal constraint set (Axis 2)
$P(x \mid \text{context}, \theta)$	Generative prior distribution over cognitive outputs (Axis 1)
ε_1	Probability threshold defining the Axis 1 feasible region
$\text{Vol}(\cdot)$	Volume measure on the Riemannian manifold M
$T_x A$	Tangent space of submanifold A at point x